Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · SIMPLE LINEAR REGRESSION

one numeric outcome, one predictor

Table 1. Regression coefficients

	(1)	**β**
(Intercept)	20.07	
	(4.54)	
	[10.88, 29.26]	
pre_score	0.64	0.81
	(0.07)	
	[0.49, 0.79]	
Num.Obs.	40	
R^2	0.66	
R^2 Adj.	0.65	
F	73.57	
RMSE	5.46	
AIC	255.36	
BIC	260.43	
Log.Lik.	−124.68	

assumption checks: linearity, normality of residuals, homoscedasticity.

Figure. Fitted-line scatter

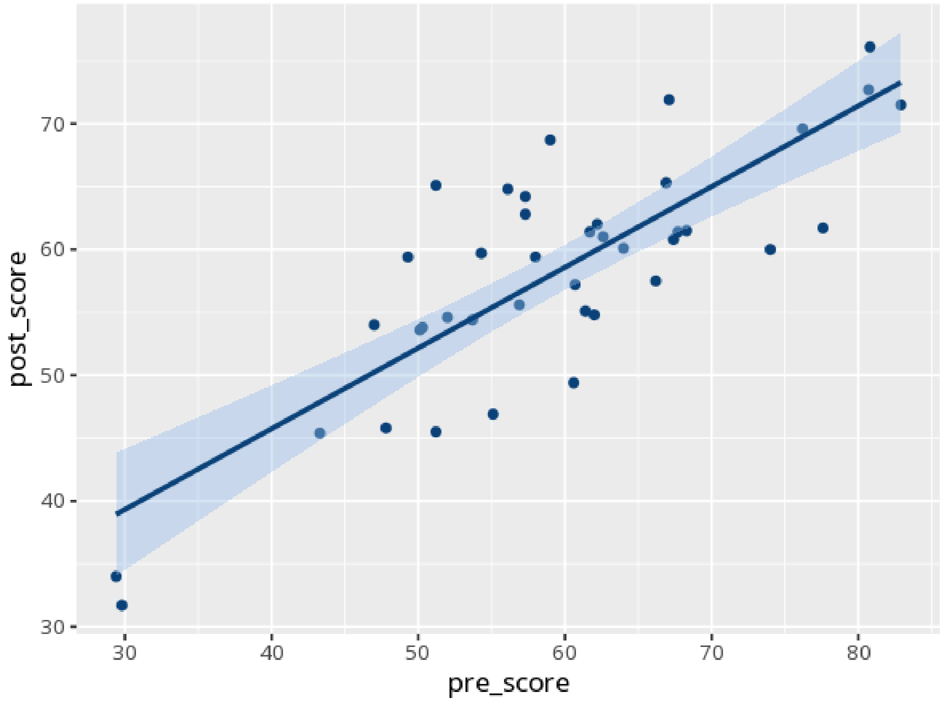
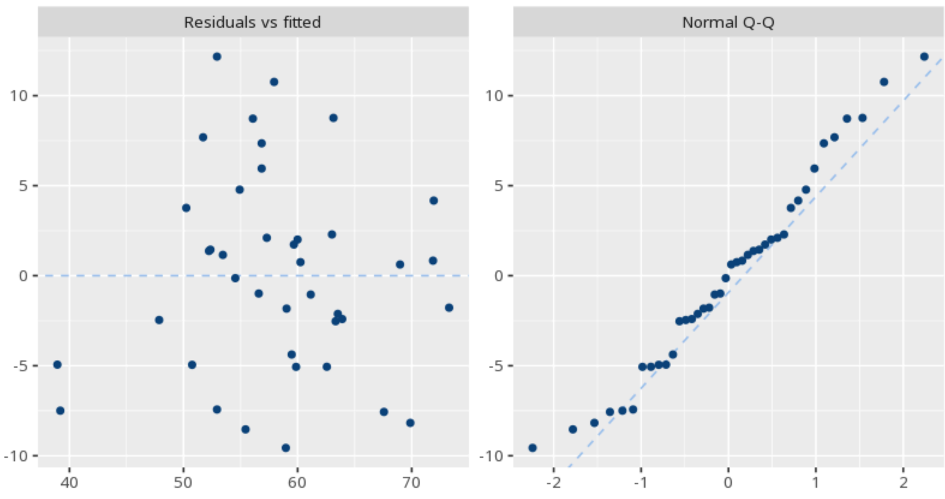


Figure. Residual diagnostics



Understanding the numbers

B. The slope of the line: the predicted change in the outcome for each one-unit increase in the predictor. Here, B = 0.64, so each one-unit increase in the predictor is associated with a change of 0.64 in the outcome.

β. The standardized slope: the change in the outcome, in standard deviations, per one SD change in the predictor - useful for comparing effect sizes on a common scale. Here, $\beta = .81$.

R^2. The share of the outcome's variance explained by the predictor. Here, $R^2 = .66$.

R^2 Adj.. R^2, adjusted for the number of predictors - the fairer figure when comparing models with a different number of predictors. Here, adjusted $R^2 = .65$.

F. The overall F-test of whether the model explains more variance than a model with no predictors at all. Here, F = 73.57.

RMSE. The root-mean-square error: the typical size of the model's prediction error, in the outcome's own units. Here, RMSE = 5.46.

AIC. The Akaike information criterion: a fit measure that penalizes model complexity, useful for comparing models fit to the same data (lower is better). Here, AIC = 255.36.

BIC. The Bayesian information criterion: like AIC but penalizes complexity more heavily as sample size grows (lower is better, for models on the same data). Here, BIC = 260.43.

Log.Lik.. The log-likelihood: how probable the observed data are under the fitted model (higher, i.e. less negative, is better); it underlies AIC and BIC. Here, Log.Lik. = −124.68.

N. The number of observations (rows of complete data) the model was fit on. Here, N = 40.

How to read this test

R^2 is the share of outcome variance explained. The predictor's B is the slope (change in outcome per unit), with p testing whether it differs from zero and the CI showing precision. β is the standardized slope (change in outcome in SDs per 1 SD change in the predictor), useful for comparing effect sizes on a common scale. Your significance threshold (α) is 0.05.

APA template: A simple linear regression gave B=0.64, t(38)=8.58, $p < .001$, R^2=.66.

Statistical basis: Ordinary least squares regression coefficient (B), SE, CI (R Core Team (2026). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.); Standardized coefficient (β) (Lüdecke D, Ben-Shachar MS, Patil I, Makowski D (2020). "Extracting, Computing and Exploring the Parameters of Statistical Models using R." Journal of Open Source Software, 5(53), 2445.). Full references in the exported CITATIONS.txt.

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